

1. Introduction

Free-radicals are typically small molecules with a minimum of one unpaired valence electron that imparts the chemical property of being highly reactive. **Reactive oxygen species (ROS)** are oxygen-containing free-radicals and **reactive nitrogen species (RNS)** are nitrogen-based free-radicals that also contain oxygen (Lander, 1997). These molecules are key signaling molecules in biological systems with effects ranging from gene expression regulation to protein activity (Ayer et al., 2014; Hancock, 2021), and are ubiquitous to aerobic systems due to respiration and photosynthesis. Intracellular sources of ROS are for example the electron transport chain in mitochondria, enzymes (e.g. NADPH oxidase, Respiratory Burst Oxidase Homologs, peroxidase), the Photosystems I and II in plant chloroplasts, peroxisomes and even intracellular bacteria. The abundance and similar valence of ROS and RNS leads to a complex interplay that has been well studied in mammalian systems and plants (Hancock, 2021; Hancock & Whiteman, 2014; Monteiro et al., 2019; Waszczak et al., 2018). Redox homeostasis is the equilibrium between oxidizing and reducing free-radicals which enables the normal cellular activities (Ayer et al., 2014; Le Gal et al., 2021). Conversely, redox disequilibrium, or **nitro-oxidative stress** (a combination of oxidative stress and nitrosative stress), alters proteins, enzymatic activity, lipids, lipoproteins and DNA, leading even to cell death (Le Gal et al., 2021).

Redox disequilibrium occurs when the organism is exposed to external stressors such as those induced by pathogen challenge, environmental contaminants, or other abiotic factors such as high temperature and high irradiance. Thus, the consequences of the Anthropogenic effects such as rising ocean temperatures and the re-emission of infrared radiations (IPCC, (Portner et al., 2022)) are a growing challenge for redox homeostasis in marine organisms. Thermal stress and high irradiance (in photosynthetic organisms) trigger the accumulation of free-radicals, which, when combined with a deficient antioxidant systems (e.g. catalase, ascorbate peroxidase, superoxide dismutase) and/or immune response are assumed to lead to cellular breakdown (Le Gal et al., 2021). One system where this redox homeostasis is particularly critical is within a **nutritional symbiosis such as reef-building corals** (cnidarian host) and their endosymbiotic partners (dinoflagellate algae of the family Symbiodiniaceae, (LaJeunesse et al., 2018)). In corals, the breakdown of symbiosis (dysbiosis) and the loss of symbiont cells leads to the host tissue becoming transparent thereby revealing the underlying white calcium carbonate skeleton, called coral bleaching (Lesser, 2011). The effects of nitro-oxidative stress are exacerbated by the symbiotic nature of reef-building corals **as not one, but two cellular systems are undergoing damage**. While corals can recover from bleaching if stress subsides and Symbiodiniaceae reinfect their tissue, coral bleaching often leads to death. Although stress-induced ROS and RNS bursts have been studied extensively in vertebrates and plants (Hancock, 2021), research on the coral holobiont (i.e. the coral host and its complex microbiome, endosymbiotic algae included) and Symbiodiniaceae is still limited (Dias et al., 2019; Gardner et al., 2017; Lesser, 1997; Roger, Russo, et al., 2022; Rosset et al., 2021). Current conservation and restoration efforts are therefore limited by our understanding of nitro-oxidative stress-induced dysbiosis.

Several mechanisms have been suggested to describe dysbiosis in corals: exocytosis of the symbiont, symbiophagy, host cell detachment (with symbiont inside), host apoptosis and autophagy (Gates et al., 1992; Oakley & Davy, 2018). However, none have been fully elucidated to date, which poses a critical challenge for the field given the increase in mass coral bleaching events (Hughes et al., 2017) and their catastrophic impacts on coral reef ecology and recruitment (Hughes, Kerry, Baird, et al., 2019; Hughes, Kerry, Connolly, et al., 2019), as well as food security (Lotze et al., 2006; Willett et al., 2019). *In vitro* research of nitro-oxidative is at its infancy in non-model species (Roger et al., 2021), such as corals. The vast majority of studies undertaken to date use *in vivo* over *in vitro* approaches due to a number of challenges associated with coral cell cultures (e.g. viability, microbial contamination, identification, proliferation, (Roger et al., 2021). Additionally, patterns and mechanisms underlying nitro-oxidative stress in live corals are masked by variability in symbiont densities (Cunning and Baker 2013), niche-specific microbial communities (van Oppen and Blackall, 2019), **and the structural complexity of the**

coral tissue and skeleton that induces microspatial differences in the diffusive exchange of ROS (Ousley et al., 2022, Wangpraseurt et al., 2022, Hughes et al., 2022). However, a recent convergence of tools has brought coral cell research to a new level with immortal cell lines generated from *Acropora tenuis* larvae (Kawamura et al., 2021), single cell RNAseq Atlas of different life stages of *Stylophora pistillata* (Levy et al., 2021), fluorescence activated cell sorting of the coral host cells and symbiotic dinoflagellates (Anthony et al., 2022; Rosental et al., 2017; Snyder et al., 2020), nanoceria antioxidant treatment of symbiotic cultures (Roger, Russo, et al., 2022), and 3D bioprinted bionic coral mimics containing live symbiotic dinoflagellate (Wangpraseurt et al., 2020; Wangpraseurt, Sun, et al., 2022). ***In light of these biological gaps and our technological advancements, our goal is to elucidate the dynamics of nitro-oxidative stress and dysbiosis in a nutritional symbiosis, from molecules to cells, and tissues to organisms, integrating recent advances in molecular and cellular physiology, nanobiotechnology and engineering.*** This work addresses both a significant biological knowledge gap in the area of basic cellular function, as well as applying a technical novel nano-bioengineering innovation of an engineered biohybrid model to the challenge of detecting spatio-temporal dynamics of free radicals within cells and their biological effects.

The proposed work is divided into three major aims (Fig.1):

AIM 1 - Quantify the ROS and RNS spatio-temporal dynamics in isolation and in symbiosis;

AIM 2 - Characterize the cellular responsiveness to controlled, targeted nitro-oxidative stress; and

AIM 3 - Manipulate and track the response to nitro-oxidative stress in a 3D bionic model.

Elucidating the Dynamics of Nitro-Oxidative Stress and Dysbiosis

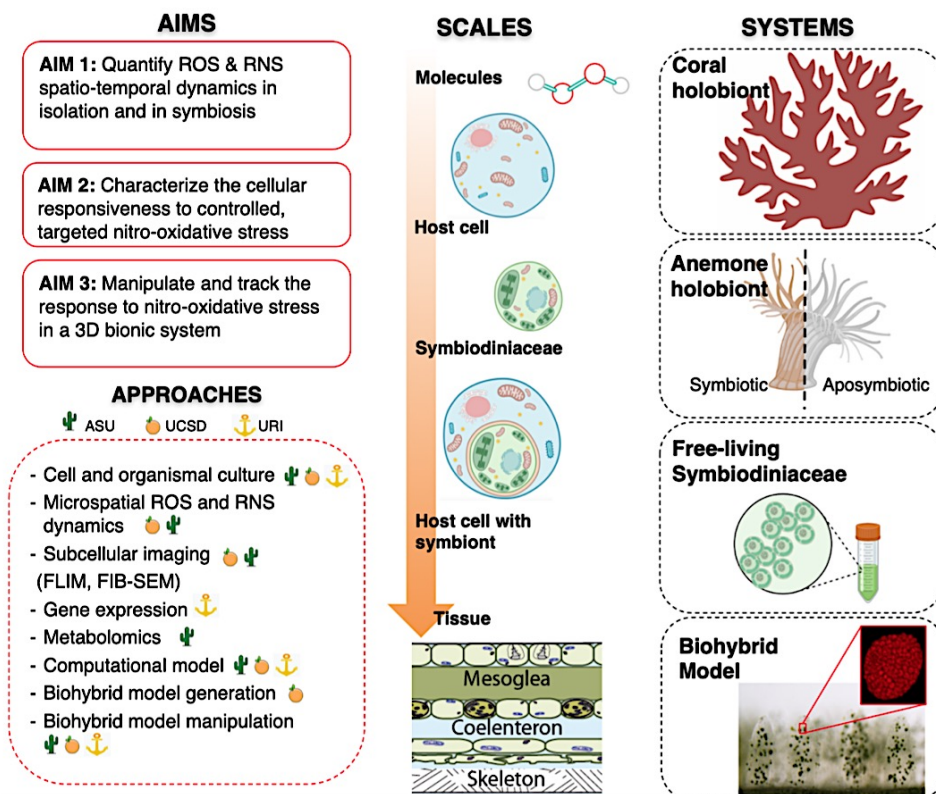


Figure 1: Proposed project overview showing the three aims, scales of research and systems used to undertake the research.

Coral reefs contribute to the well-being, culture, economy, global trade and coastal protection of over 100 nations. Record-breaking climatic events are threatening coral reef survival and human safety at

an unprecedented rate. Under business-as-usual practices, by 2050 we are expected to lose up to 90% of tropical and subtropical reefs worldwide, with devastating effects on global biodiversity, coastal communities stability and food security (Portner et al., 2022). The symbiotic relationship between the coral host and its endosymbiont is central to the survival and growth of coral reefs worldwide (Roth, 2014). ***The convergence of disciplines proposed here will allow us to characterize the complex dynamics at play during stress driving the nutritional dysbiosis that leads to coral bleaching and cell death through multi-scale manipulation of isolated cells, symbiotic cells, and a 3D biohybrid coral host model.*** We will start by quantifying and mapping the different types of free-radicals (reactive oxygen species, ROS, and reactive nitrogen species RNS) and their dynamics *in vivo* and *in vitro* using reef-building corals (colonies and dissociated cells), sea anemones (single individuals and dissociated cells), and free-living symbiotic dinoflagellate algae (cultured cells), and in a biohybrid coral system in ***Aim 1*** (Fig.1). We will then apply typical stressors to the different systems aforementioned to quantify and map cnidarian and dinoflagellate cell-to-cell interactions under nitro-oxidative stress in ***Aim 2*** (Fig.1). The biohybrid system provides a tractable system to build the cellular response back into the organismal 3D architecture and spatially map free-radical diffusion, we will manipulate the impact of tissue architecture and cell density patterns on free-radical dynamics to understand the 3D system in ***Aim 3*** (Fig.1). The combined *in vivo*, *in vitro* and biohybrid studies will all be integrated with cellular, metabolomic, gene expression, and physiological assessments to understand the molecular cascades and cellular functions influenced by free-radical stress in a complex symbiotic system. ***This multi-scale and transdisciplinary approach is strategically chosen to generate a comprehensive, spatio-temporal understanding of nitro-oxidative stress in dysbiosis, which will help towards improving our understanding of the tipping points between redox homeostasis and imbalance, as well as informing applications to reduce coral reef loss and improve restoration.***

2. Background

Symbiosis between multiple organisms results in complex interactions with emergent properties that can have substantial implications for environmental resistance and resilience (Douglas, 2010). In particular, nutritional symbioses, which include the exchange of metabolic products between cells of differing taxa, present a case of tight metabolic integration, with the potential for devastating consequences of dysbiosis. An ideal model for testing symbiotic communication, nutritional exchange, and immunity is that of the symbiosis between cnidarians and their endosymbiotic dinoflagellates in the family Symbiodinaceae. Translocation of metabolic products from algal photosynthesis can provide close to 100% of the daily carbon requirement for maintenance respiration of the coral host (Falkowski et al., 1984). This metabolic exchange provides the fuel for coral calcification, which forms the structural basis of the entire ecosystem. The coral-Symbiodinaceae symbiosis is fragile, with documented linkages to oxidative stress driven dysbiosis under increasing temperatures and high light levels resulting in loss of the symbiotic cells (i.e., coral bleaching, (Lesser, 2011)), and consequently an energetic deficit (Anthony et al., 2009; Falkowski et al., 1984) and often mass mortality (Hughes et al., 2017). Therefore symbiotic metabolic crosstalk is critical to maintenance of the symbiosis and environmental stress tolerance, but the detailed cascade and individual and combined roles of ROS and RNS are not fully clear.

Nitro-oxidative stress is a primary driver of symbiotic dysbiosis

The dysfunctions brought on by nitro-oxidative stress are particularly problematic in the case of symbiotic organisms considering the “cell within a cell” conformity: if either partner were to undergo dramatic homeostatic changes, both partners will be impacted. At lower abundances of ROS and RNS, positive signaling processes are activators for regulatory feedback (Fig. 2A), but as ROS and RNS increase as a function of cellular metabolism at higher light and temperatures, the damaging effects of protein misfolding, loss of mitochondrial function and PSII function can tip a cell from redox homeostasis to accumulating damage and cell death (Fig. 2A). Mitochondria are known to suffer from oxidative stress (Youle & van der Bliek, 2012) and within the symbiotic relationship, this means host mitochondria and

symbiont mitochondria are at risk of mitophagy and fission dysfunctions (Fig. 2A). Furthermore, in a photosynthetic endosymbiont, the Photosystem I and II are considerable producers of free-radicals under stress-free (photosynthesis) and stressful conditions (high irradiance) (Foyer, 1996; Szabó et al., 2020; Trubitsin et al., 2014) (Fig. 2A).

The complex dynamics of nitro-oxidative stress are due to **signal detection and the interplay** of a number of ROS and RNS that can react with each other (Fig. 2B), potentially leading to a misinterpretation of the drivers and response. For example, in this complex cascade, oxygen (O_2) can react to become hydrogen peroxide (H_2O_2) and in the absence of enough of the antioxidant superoxide dismutase (SOD), can be converted to the even more damaging hydroxyl radical ($\cdot OH$). Additionally, oxygen (O_2) can react with nitric oxide synthase (NOS) to form the RNS of nitric oxide ($NO\cdot$) and peroxynitrite ($ONOO^-$), with superoxide ($O_2^{\cdot -}$) linking the ROS and RNS pathways. Given that the lifespan of ROS and RNS can be as short as a few μ seconds (Diaz & Plummer, 2018), a reduction in superoxide ($O_2^{\cdot -}$) could be interpreted as a reduction in ROS, if the levels of RNS, especially peroxynitrite ($ONOO^-$), are not monitored in conjunction. Indeed, the formation of $ONOO^-$ from $O_2^{\cdot -}$ and $NO\cdot$ has a reaction rate constant one order of magnitude higher than that of the formation of H_2O_2 from $O_2^{\cdot -}$ catalyzed by superoxide dismutase (SOD), **thus it is essential to quantify both $ONOO^-$ and $NO\cdot$** (Fig. 2B). Currently, the diversity and interplay of ROS and RNS, and the **lack of molecule-specific signal detection methods hampers our understanding of the spatio-temporal drivers and cascades of nitro-oxidative stress driven dysbiosis.**

Multiple systems for a multiscale approach

In order to disentangle the roles of ROS and RNS in dysbiosis, we propose to work across scales from isolated cells, symbiotic cells, tissues, organisms, to a 3D bioprinted coral model. Such a multidisciplinary and multiscale project has previously been limited by technical and technological challenges. In recent years, our contributions to the advancement of *in vitro* coral cell research (Roger, Darko, et al., 2022; Roger et al., 2021; Roger, Russo, et al., 2022) and 3D bioprinting using marine organisms (Martin et al., 2021; Wangpraseurt et al., 2020; Wangpraseurt, You, et al., 2022) have provide the essential foundation for this work. The development of coral cell culture methods and 3D bioprinting now enable us to unlock a new level of cutting edge symbiosis research that represents a missing link between *in vivo* and *in vitro* approaches. The systems we will use here, namely *in vitro* and *in vivo* reef-building corals, *in vitro* and *in vivo* model anemone (*Exaiptasia*), free-living Symbiodiniaceae and the biohybrid coral (Fig. 1), represent the symbiotic relationship between coral cells and dinoflagellate algae, which is the cornerstone of the productivity of the coral holobiont that acts as the engineer for the entire coral reef ecosystem. As such, **these systems represent ideal and critical focal models for advancing our understanding of cellular redox homeostasis during nitro-oxidative stress.**

Applicability to integrative research in biology solicitation

Studying symbiosis presents the exemplar technical challenge of biological integration. There is both the need for separating the signals or responses of one partner from the other to understand how they interact, as well as their responses in combination. The high reactivity of ROS and RNS, and their interplay described above, makes for considerable technical measurement and integration challenges. The combination of disciplines proposed here (molecular sciences, genomics, biochemistry, and nanoengineering) will enable us to study the different parts of the symbiotic complex (host cell and

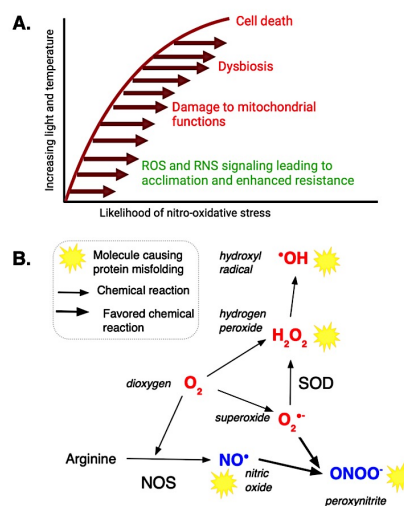


Figure 2: Schematic representation of the likelihood of nitro-oxidative stress with abiotic stressors (A) and the interactions between certain reactive oxygen species and reactive nitrogen species (B)

endosymbiotic algae) in isolation and combined under different scenarios to understand how and when this symbiosis shifts from beneficial nutrient exchange to dysbiosis (Bénard et al., 2020) and to disentangle the roles of ROS and RNS. The use of the recently developed biohybrid coral system enables the groundbreaking capacity for flexibility of tissue design such that the overall construct shape (bioink mimicking “tissue”) can be used for manipulating cell density, cell type, 3D tissue architecture, bioink composition and culture conditions while also being based on a realistic supporting material (mimicking coral skeleton) (Wangpraseurt et al., 2020).

3. Intellectual Merit

The proposed research brings together a synergistic research team combining marine biology, molecular sciences, ecophysiology, nanoengineering and 3D biofabrication to develop a detailed and integrative understanding of the cascade of nitro-oxidative stress in host and symbiont cells, both in isolation and in symbiosis, and also embedded in bioengineered tissues. This work harnesses the capabilities of advances in coral cell culture in conjunction with a recently developed coral biohybrid model and the flexibility of a non-calcifying symbiotic and aposymbiotic cnidarian model system (*Exaiptasia*) (Weis et al., 2008). Our toolkit of ROS and RNS spatiotemporal characterization, combined with metabolomics, gene expression, and physiology, will allow us to quantify free-radical signals (ROS and RNS) and characterize free-radical stress (biological response to ROS and RNS) from molecule to cell, cell to tissue, and tissue to organism, in the context of the 3D micro-landscape and architecture of complex tissues. Our findings will help describe the interactions between different types of free-radicals (reactive oxygen species, ROS, and reactive nitrogen species, RNS), the first signs of dysbiosis at different scales, and the role of tissue patterns and architecture in modulating the organismal response to nitro-oxidative stress. The successful completion of this project will generate a cnidarian-dinoflagellate model specific understanding of coral bleaching and dysbiosis which will open new possibilities for coral reef conservation, restoration and rehabilitation. Further, the information gained from measurements of isolated cells will be broadly applicable to cellular redox homeostasis across taxa from plants to mammals. Notably, we will apply our novel 3D bionic coral hybrid model to generate innovative advancements in bioengineering for non-model systems, with direct applications for the use of nanotechnology in marine ecology and restoration.

4. Integrative and Interdisciplinary Team

This synergistic project leverages our team’s (Fig. 3) expertise in molecular biology and cnidarian cell culture (Arizona State University, ASU); nanoengineering and 3D biofabrication (University of California San Diego, UCSD); and molecular and organismal ecophysiology (University of Rhode Island, URI) to elucidate nitro-oxidative stress response. **Roger** (ASU) is a marine biologist and molecular scientist with an extensive multidisciplinary background ranging from geochemistry and biomineralization to *in vitro* cnidarian cellular biology (reef-building corals and sea anemones), to applied chemical engineering, nanomedicine and microfluidics. Roger’s capacity for complex imaging (Roger, Adarkwa Darko, et al., 2022), cell culture in non-model systems (Roger et al., 2021), and chemical engineering (Roger, Russo, et al., 2022) provides the critical assessment of reactive oxygen and nitrogen species. Roger will be responsible for *in vitro* and *in vivo* quantification and imaging of ROS and RNS, and associated metabolomics. As lead-PI, Roger will be responsible for coordinating multidisciplinary research integration, all team efforts (all-hands meetings, timelines) and monitoring key performance indicators (KPIs). **Wangpraseurt** (UCSD) is an interdisciplinary marine biologist and engineer with research foci at the interface between marine biology, optics and (bio)engineering. Wangpraseurt is an early career leader in the field of nanoengineering and 3D biofabrication for symbiosis (Martin et al., 2021), which has enabled the generation of the coral biohybrid model (Wangpraseurt et al., 2020;

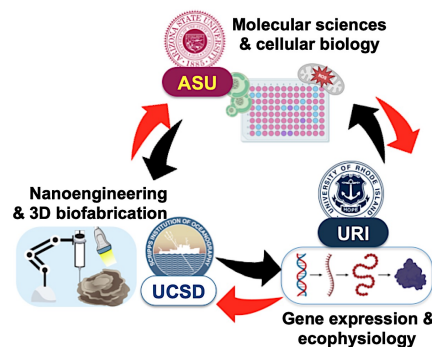


Figure 3: Our interdisciplinary team at a glance.

Wangpraseurt, You, et al., 2022) we will employ here to manipulate cell density and patterns in a controlled 3D environment. Wangpraseurt will lead the UCSD experimental work on the 3D biofabrication and manipulation of the bionic hybrid model and supervise the UCSD postdoc and team in analysis, writing, and presentation of experimental work. He will also contribute to the multidisciplinary and multi-scale integration of data, participant training, data management, and science education through Broader Impacts. **Putnam** (URI) is a molecular eco-physiologist and has published extensively at the intersection of marine ecophysiology, epigenetics, symbiosis, and climate change. Putnam's extensive background in coral biology (Putnam et al., 2017), molecular ecophysiology (Putnam et al., 2016), and multi-omics (Williams, Pathmanathan, et al., 2021) provides the foundational context for responsiveness of the coral-dinoflagellate symbiosis to nitro-oxidative stress. Putnam will be responsible for experiments and physiological and transcriptomic characterization of samples, will lead and supervise writing and presentation of experimental results and Broader Impacts at URI, and will work with all PIs for integration of data across scales, participant training, data management, and postdoc and student mentoring.

5. Research Plan

Study Species: The study species we have chosen here are: (1) the Indo Pacific coral *Pocillopora acuta* (genome (Stephens et al., 2022)); (2) the anemone *Exaiptasia diaphana* found globally (genome (Baumgarten et al., 2015)); (3) Symbiodiniaceae cultures such as *Brevium minutum* (Roger, Russo, et al., 2022), genome (Shoguchi et al., 2013)); and (4) a 3D printed bionic coral model capable of growing Symbiodiniaceae within bioink on skeletal mimics (Wangpraseurt et al., 2020; Wangpraseurt, Sun, et al., 2022). Both *P. acuta* and *E. diaphana* are cultured in the Roger and Putnam labs and Wangpraseurt is the pioneer of the 3D printed bionic coral model (Fig. 4). Our extensive publication record and constant supply of material supports the likelihood of project success (Putnam et al., 2013, 2020; Roger et al., 2021; Roger, Russo, et al., 2022; Wangpraseurt et al., 2020; Wangpraseurt, Sun, et al., 2022; Williams, Chiles, et al., 2021).

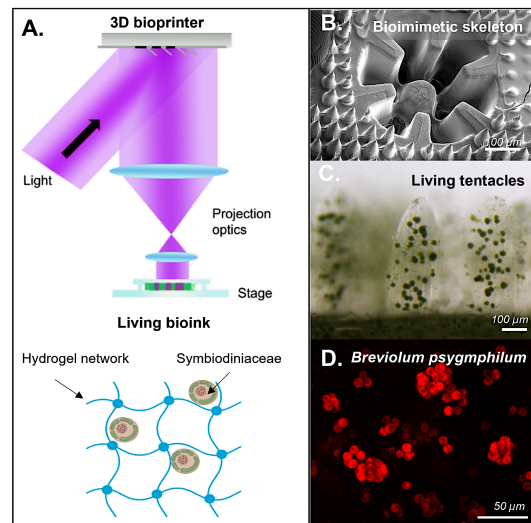


Figure 4. Biohybrid model fabrication. A. Our 3D bioprinter set-up and engineered hydrogels with encapsulated Symbiodiniaceae. B.-D. Successful biofabrication of coral-algal model. Wangpraseurt et al. 2020 & 2022.

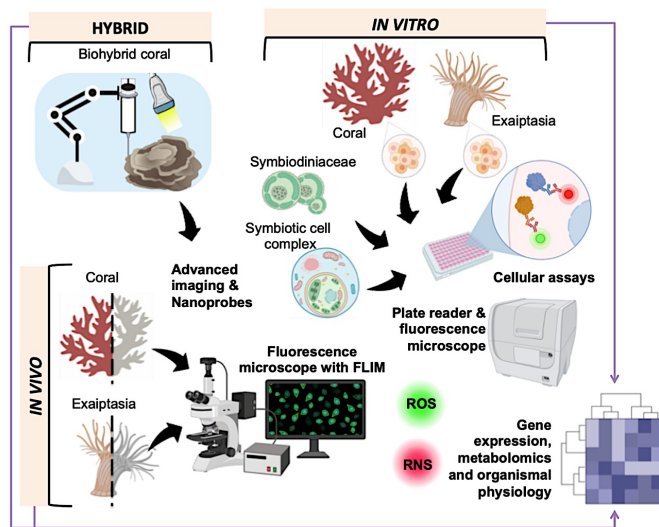


Figure 5: Aim 1 approach graphic representation

Aim 1: Quantify the ROS and RNS spatio-temporal dynamics in isolation and in symbiosis

Rationale 1: Symbiotic organisms, especially nutritional symbiosis, provide an ideal study system to quantify the spatio-temporal dynamics of ROS and RNS production. Establishing a baseline of ROS and RNS production in isolation will allow for a comparison of individual dynamics under stress and how this may deviate from that assayed *in hospite*. The first step of our work is therefore to address the following question: **How does intracellular ROS and RNS regulation vary between cells in free-living, symbiotic, and aposymbiotic states?** For this, we will: **Characterize the ROS and RNS**

spatio-temporal dynamics and biological response in cnidarian cells and symbiont *in vitro* (Aim 1.1); *in vivo* (Aim 1.2); and in a hybrid system (Aim 1.3) (Fig. 5).

Approach 1: We will first generate the cnidarian host cells and symbiotic algae that will be studied *in vitro*, *in vivo* and in a hybrid system.

Aim 1.1: *In vitro* testing will be performed by dissociating cells from whole organisms (Roger et al., 2021), or **starting Symbiodiniaceae cultures from cell cultures** (Roger, Russo, et al., 2022) commercially available (e.g., National Center for Marine Algae and Microbiota, NCMA, Bigelow Labs). We will first quantify ROS and RNS concentrations under ambient cell culture conditions (Roger et al., 2021) using fluorescent markers and reagents such as CellROX (flow cytometry, *sensu* (Dungan et al., 2022)), and DCFH-DA, DAF2-DA following methods developed by Roger using a plate reader (Roger, Russo, et al., 2022). The emphasis will be on characterizing the interplay between ROS and RNS. Our recent tests focusing on H₂O₂ and NO measurements in parallel experiments using free-living symbiotic algae showed an absence of significant NO increase under acute thermal stress (+6 °C) whereas H₂O₂ concentration significantly increased (Fig 6). The change in NO concentration can be masked due to instrument limitations; only one type of fluorescence signal could be measured at one time (Roger, Russo, et al.,

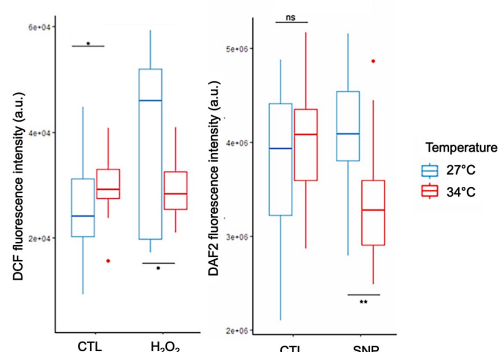


Figure 6: H₂O₂ and NO concentration relative to DCF (reacted DCFH-DA) and DAF2 (reacted DAF2-DA) under control conditions and acute thermal stress (+6°C) for 1 hour in free-living symbiotic algae *Breviolum minutum* (Roger et al., 2022)

2022). In the same study, positive controls (for ROS: H₂O₂, for RNS: Sodium nitroprusside SNP) showed a significant decrease under acute thermal stress during the 1h exposure period suggesting an efficient response mechanism under these conditions (Fig 6). To address potential limitations, we will combine the well established plate reader assays with advanced confocal fluorescence imaging commonly used by PI Roger for **Fluorescence Lifetime Imaging (FLIM)** (conducted routinely at ASU's Biosciences Core facilities). FLIM will help determine the spatio-temporal changes of ROS and RNS concentration by recording changes in fluorescence over time in regions of interest inside the cells (fluorescence time-series using AND-logic based fluorescent probes (Wang et al., 2013; Wu et al., 2019)).

To quantify ROS and RNS we will assess changes in concentration relative to fluorescence intensity using probe combinations for plate reader and confocal fluorescence microscopy. **AND-logic based probes are tailored to produce a fluorescent response after the simultaneous or sequential reaction of two or more highly reactive molecules** (Wu et al., 2019). Standards are run at the ASU core facility, therefore absolute and relative values will be available for analysis. Cellular assays will be conducted in 96 well-plates (1x10⁴ cells^{mL}) to quantify response to free-radical species levels (Table 1). Analysis of the free radical and cellular measurements will be conducted with repeated measures ANOVA.

Table 1. Assays for free radical species generation and cellular response to free radical species

Free-radicals	Anticipated Response	Measurement method	Reference
H ₂ O ₂	Initial increase then decrease due to antioxidant response	DCFH-DA (plate reader), AND-logic based probe (FLIM), colorimetric Ampliflu Red (plate reader)	(Roger, Russo, et al., 2022; Wangpraseurt, Sun, et al., 2022; Wu et al., 2019)
ONOO ⁻	Increase linked to NO [•] and O ₂ ^{•-} burst	DAF2-DA (plate reader), AND-logic based probes (FLIM)	(Roger, Russo, et al., 2022; Wu et al., 2019)
NO [•]	No increase with thermal stress reduction with ONOO ⁻ formation	DAF2-DA (plate reader), AND-logic based probes (FLIM)	(Roger, Russo, et al., 2022; Wu et al., 2019)
[•] OH	Trend shadowing that of H ₂ O ₂	DCFH-DA(plate reader), AND-logic based probes (FLIM)	(Roger, Russo, et al., 2022; Wu et al., 2019)
O ₂ ^{•-}	Highly dynamic due to the formation of H ₂ O ₂ and ONOO ⁻	DCFH-DA(plate reader), AND-logic based probes (FLIM)	(Roger, Russo, et al., 2022; Wu et al., 2019)
Biological Response	Cellular Assays	Measurement method	Reference
Mitochondrial integrity	Mitotracker fluorescent dye	Confocal microscopy	(Dunn et al., 2012)
Antioxidant response	Catalase, superoxide dismutase	Microplate assay on cell culture or	(Krueger et al., 2015)

	(protein normalized)	tissue lysate	
Programmed cell death	caspase	Microplate assay on cell culture or tissue lysate	(Tchernov et al., 2011)
Cellular defense and repair	NADPH oxidase, NOX	Microplate assay on cell culture or tissue lysate	(Zielonka et al., 2017)
Metabolism	Respiration and photosynthesis	Oxygen respirometry	(Putnam & Gates, 2015)

At the cell culture level, replicate measurements of cellular respiration and photosynthesis will be taken in the dark and light, respectively, using a 24 well-plate (80µl well volume) Presens Sensor Dish Reader in culture media conditions in benchtop incubators. Replicate samples (n=4 per treatment) will also be collected from each culture or mixed symbiotic cell sample for gene expression quantification using TagSeq (Lohman et al., 2016). Statistical analysis of the physiological measurements will be conducted with ANOVA for the host, symbiont, and combined cells. Gene expression will be quantified as is common in Putnam group (Chille et al., 2021; Chille et al., 2022) using DESeq2 (Love et al., 2014) to test for differences between treatment groups and Weighted Gene Coexpression Network Analysis (WGCNA (Langfelder & Horvath, 2008)) to examine expression modules and their correlation with stress state (ROS and RNS), stress response (mitochondrial membrane potential, catalase superoxide dismutase, caspase, and NADPH oxidase), as well as organismal measures (respiration and photosynthesis). Samples preserved at -80°C for metabolomics will be processed following a water-soluble polar metabolite-adapted protocol as described in (Williams, Chiles, et al., 2021), followed by analysis with liquid chromatography-mass spectrometry.

Aim 1.2: *In vivo* testing will be performed on coral polyps and anemones using z-stacking and confocal fluorescence microscopy with FLIM. While corals are less biologically stable when bleached (i.e. death can occur rapidly), model anemone *E. diaphana* can be maintained in an aposymbiotic state for extended periods of time. This will enable us to image the ROS and RNS dynamics in a cnidarian host *in hospite* in the absence of symbionts using the Fluorescence Lifetime Imaging (FLIM) approach. We will also complete these measurements in bleached corals (*sensu* (Huffmyer et al., 2020)). At the organismal level, replicate measurements of cellular respiration and photosynthesis will be taken in the dark and light, respectively, using a 10-channel temperature compensated Presens Oxy10 ST (G2) to measure oxygen change through time in sealed stirring chambers. Replicate samples (n=4 per treatment) will also be collected from each culture or mixed symbiotic cell sample for gene expression quantification using TagSeq (Lohman et al., 2016). Gene expression and its connection to cellular and physiological state will be quantified and analyzed as described in Aim 1.1.

Aim 1.3: The 3D biohybrid model approach will consist of embedding cells in a 3D bioprinted construct, the biohybrid coral (Wangpraseurt et al., 2020), and measuring the concentration of ROS and RNS within the cells and the bioink “tissue” (Wangpraseurt, Sun, et al., 2022). The 3D biohybrid model will be generated according to the methods of (Wangpraseurt et al., 2020; Wangpraseurt, Sun, et al., 2022), which include a 2-step continuous light projection-based bioprinting of photopolymerizable alginate-methacrylate (AlgMA) hydrogel and cellulose-derived nanocrystals (CNC), with the Symbiodiniaceae and/or cultured coral cells mixed within the ink. This construct is printed onto a Pocilloporid mimic skeleton, which itself is 3D printed with a polyethylene glycol diacrylate-based polymer (PEGDA) doped with CNC. ROS and RNS within the biohybrid tissue will be quantified as described in Aim 1.1. Further the cells will be extracted from the bioink tissue using established protocols (via alginate lyase (Blanco-Cabra et al., 2020)) following exposure to ambient conditions and cellular assays, metabolomics and gene expression will be completed as described in Aim 1.1. The success of our 3D bioprinted model will be evaluated using our well established approaches (Wangpraseurt et al., 2017, 2020; Wangpraseurt, You, et al., 2022). To validate the spatial accuracy of our 3D bioprinting approach at least five replicate coral tissue and skeleton models will be imaged using optical coherence tomography (Wangpraseurt et al., 2017). Additionally, at least five skeleton models will be imaged via environmental Scanning Electron Microscopy (SEM) (Wangpraseurt et al., 2020). The mechanical properties of the created hydrogel

scaffolds will be characterized. The Young's modulus will be evaluated for at least five bioprinted scaffolds by performing stress-strain measurements with a microscale mechanical strength tester (Microsquisher, CellScale; (Wangpraseurt et al., 2020)), with calculations performed in the linear region of the stress–strain curve.

Anticipated Outcomes 1: Collectively, our detailed characterization of ROS and RNS will provide the first such simultaneous quantification of these free radicals in both isolated and symbiotic cells and bioprinted constructs. Due to the complexity of a full symbiotic system at the organismal level, our approach reduces the risks and improves our capacity for success of measurements by building on our existing experimental and methodological expertise (e.g., (Putnam et al., 2013; Roger, Russo, et al., 2022; Wangpraseurt et al., 2020; Wangpraseurt, Sun, et al., 2022)). Additionally, our choices of multiple model systems provide alternative approaches if, for example, the calcified skeleton interferes with the assays.

Aim 2. Characterize the cellular responsiveness to controlled, targeted nitro-oxidative stress

Rationale 2: Outcomes of dysbiosis due to nitro-oxidative stress include exocytosis of the symbiont, symbiophagy, host cell detachment, host apoptosis and autophagy (Gates et al., 1992; Oakley & Davy, 2018). All of these processes involve the general cascade of oxidative stress with excess excitation energy damage to PSII and the D1 protein leading to these various forms of dysbiosis (Lesser, 2011; Oakley & Davy, 2018; Venn et al., 2008). However, the specific temporal cascade involving nitro-oxidative stress (Gardner et al., 2017) is still lacking key details of the ROS RNS interactions and quantification, and the spatial patterns of ROS and RNS with respect to epidermis, gastrodermis, and cell types are not yet available. It is therefore critical to understand how, when, and where symbiosis shifts from beneficial to detrimental and answer the following question: *What is the cascade of dysbiosis due to ROS and RNS across the cellular level and the tissue levels?* For this, the following subaims have been identified: **(Aim 2.1) Describe the response of subcellular compartments and organelles to nitro-oxidative stress;** **(Aim 2.2) Characterize the metabolomic, transcriptional and physiological responses to nitro-oxidative stress;** and **(Aim 2.3) Conduct computational modeling of antioxidant response and redox homeostasis tipping points (Fig. 7).**

Approach 2: The experimental design we will use for Aim 2 (Table 2) is a series of treatments including: 1) an ambient control of ambient media and culture conditions for the cell and organismal levels, 2) exposure of samples to thermal stress alone (+6 °C), 3) high light stress alone (ambient irradiance x5, e.g., for cultures $\sim 300 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$ at ambient and $\sim 1500 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$ at high), 4) the combination of high temperature and light (+6 °C and $\sim 1500 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$); 5) a positive control for ROS hydrogen

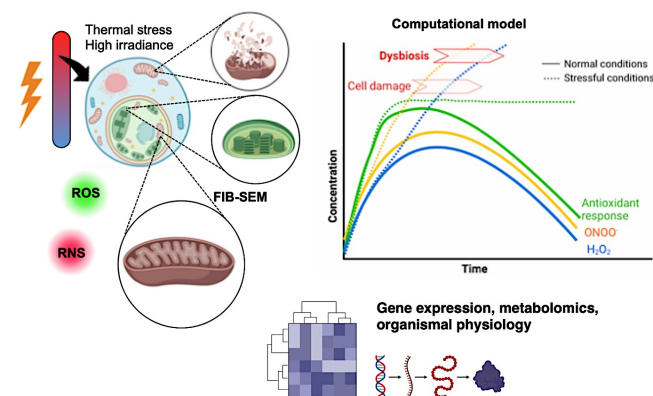


Figure 7: Aim 2 approach graphic representation

peroxide (H_2O_2 ; 100 μM); 6) a positive control for RNS sodium nitroprusside (SNP; 1 μM); and 7) the combination of ROS and RNS (100 μM H_2O_2 and 1 μM SNP). For each of these, the experiments will be conducted as a timeseries, with T0, +1h, +8h, +24h, and +48h. Experiments for Aim 2 will be performed using all levels of our model systems (Fig. 1) beginning with our cell scale and working stepwise to higher levels. Given that the cell cultures (Roger et al., 2021) and biohybrid model (Wangpraseurt, Sun, et al., 2022) are in culture media and the coral and anemone models are in seawater, we have shown it is possible to easily control the environmental conditions with media or seawater additions, or changing the light and temperature in benchtop incubators (e.g., (Roger, Russo, et al., 2022; Wangpraseurt, Sun, et al., 2022)).

Table 2. Proposed experiments to address ROS and RNS generation or addition and biological response in the presence and absence of external stressors and in isolation and combination.

Treatment	Source of stress	Expectation
Control	NA - Ambient conditions	Control for comparison of treatments
High temperature	Externally mediated metabolic processes exacerbate ROS and RNS via metabolism then photosynthesis	Enhanced rates of reactions will ROS and RNS with detrimental biological responses
High light	Externally mediated metabolic processes exacerbate ROS and RNS via photosynthesis then metabolism	Enhanced rates of reactions will ROS and RNS with detrimental biological responses. Light exposure in the absence of temperature allows the data to be interpreted with respect primarily to the cause and effects on the photosystems
high temperature and light	Externally mediated metabolic processes exacerbate ROS and RNS via photosynthesis then metabolism	Spatial and temporal differences due to interacting stressors will highlight the ROS and RNS crosstalk within isolated cells and symbiotic partners
ROS	External ROS (H ₂ O ₂)	External addition of H ₂ O ₂ isolates free radical responses to ROS mediated effects in the absence of thermal effects on biology and biochemistry
RNS	External RNS (SNP)	External addition of SNP isolates free radical responses to RNS mediated effects in the absence of thermal effects on biology and biochemistry
ROS and RNS	External ROS RNS (H ₂ O ₂ , SNP)	Expect to see a synergistic effect of simultaneous exposure to ROS and RNS and can disentangle this in the absence of thermal effects on biology and biochemistry

Aim 2.1: To observe the response of subcellular architecture compartments and organelles to nitro-oxidative stress we will use FIB-SEM (FIB-SEM, Focused Ion Beam-Scanning Electron Microscopy) (Uwizeye et al., 2021) to map and assess the condition of different organelles to nitro-oxidative stress in across scale in our model systems. Sample preparation consists in gold sputter coating followed by sample trimming with a 90° diamond knife (Diatome) to expose cells at two surfaces (the imaging surface and the surface perpendicular to the FIB). Imaging will be performed using routine settings at regions of interest. Data will be recorded using commercially available software (Fibics Inc. and Carl Zeiss Microscopy GmbH) and post-processed for subcellular morphometrics analysis using plugins for Fiji and 3D slicer (3D slicer computing platform: Slicer) (see (Uwizeye et al., 2021)). This approach will generate 3D views of organelles of the different cells studied under different levels of nitro-oxidative stress to assess physical damage leading to dysbiosis. FIB-SEM imaging will be performed at the microscope core unit at ASU and/or UCSD.

Aim 2.2: In order to elucidate the pathways triggered by and responding to nitro-oxidative stress we will quantify the sample gene expression, metabolomic profiles, and rates of respiration and photosynthesis under the treatments listed above. First we will test for broad correlations between gene expression and metabolites (e.g., (Williams, Chiles, et al., 2021; Williams, Pathmanathan, et al., 2021)). Second we will take a targeted spatial approach based on specific dissections of tissues or localization within the 3D printed construct (Wangpraseurt, Sun, et al., 2022) based on ROS and RNS measurements. We will test to WGCNA module-module correlations between expression and metabolites, as well as gene ontology and gene set enrichment analyses combined with metabolite pathway enrichment analyses to determine molecular signatures of up/down regulation during nitro-oxidative stress triggered by different abiotic factors (e.g., temp, light, temp+light, external ROS and RNS). We will also examine correlations between eigengene and eigenmetabolite values with cellular respiration and symbiont photosynthetic rates and cellular responses defined in Aim 1.

Aim 2.3: To define the boundaries of the antioxidant defense and tipping points we will design a computational model of stress-induced dysbiosis. Computational modeling has been used extensively to describe wound healing rates and stress response (Reynolds et al., 2006; Segal et al., 2012; Torres et al., 2019): the computational model will consist in a number of nonlinear differential equations modeling the concentration of ROS and RNS (H₂O₂, NO[•], ONOO⁻, •OH) over time depending on growth conditions (e.g. temperature, irradiance). Basic interactions between the variables aforementioned integrated with subcellular responses (mitochondrial response, programmed cell death, dysbiosis) will be predicted using

Latin Hypercube sampling and parameters estimation methods (Renolds, Segal, Roger et al. unpublished data; (Torres et al., 2019)). The model predicted response will be compared to *in vitro*, *in vivo* and biohybrid findings to assess parametrization success and define equilibrium and tipping points of symbiosis.

Anticipated Outcomes 2: This detailed time series of multiple free radical species with high spatial and temporal resolution will be the most advanced quantification for the field to date. Further, the use of treatments with light and temperature driving increases in ROS and RNS in comparison to the additions of external ROS and RNS provides a powerful approach to teasing apart thermal and excess excitation energy effects from the direct effects of free radicals. If the computational model is failing to capture the dynamics seen in the measured time series, we could also assess the linkages between primary drivers and biological responses using a model selection approach (e.g., AIC, BIC) or structural equation modeling. While single cell RNASeq is at the cutting edge, to date there is only one publication for this in reef building corals and the cost for the throughput of our assays would be prohibitively high. Therefore we have proposed the well documented and high resolution TagSeq approach.

Aim 3. Manipulate and track the response to nitro-oxidative stress in a 3D bioprinted coral model

Rationale 3: Despite their simple body plan, cnidarians are composed of multiple tissue layers with different functions, not all engaged in symbiosis with Symbiodiniaceae, but all can be impacted by nitro-oxidative stress. Hydrogen peroxide, for example, is a free-radical with a relatively long residence time compared to the hydroxyl radical (hours to day vs μsec , (Diaz & Plummer, 2018) and translocates easily across lipid membranes due to its lack of charge (Hancock, 2021). To fully understand the organisms response to nitro-oxidative stress, we have to consider the heterogeneous microenvironmental landscape of the coral host (Hughes et al., 2022), which is affected by multiple tissue layers of different thicknesses, complex skeletal morphologies, and the build-up of fluid-flow driven diffusive boundary layers (Kühl et al., 1995; Ricci et al., 2022). We will thus characterize nitro-oxidative dynamics in a realistic 3D biohybrid coral model system to systematically test how coral tissue type and morphology as well as skeletal architecture modulate nitro-oxidative stress. For this, the following subaims have been identified: **(Aim 3.1) characterize the response to nitro-oxidative stress in tissues of different architectures (tentacles vs. connective tissue, thick vs. thin tissues and oral vs. aboral epidermis); and (Aim 3.2) Characterize the response to nitro-oxidative stress in tissues with different symbiont densities. (Aim 3.3) Characterize the response to nitro-oxidative stress in tissues with different skeletal porosities and microarchitectures (Fig. 8).**

Approach 3. To systematically evaluate the role of the 3D coral landscape in affecting nitro-oxidative stress, we will engineer a range of biohybrid coral host models using our established 3D bioprinting platform. The different models will be tested for ROS and RNS dynamics (as described above) under control conditions (see above, 25°C and downwelling irradiance= $300 \mu\text{mol photons m}^{-2} \text{s}^{-1}$, $n=6$ bionic corals per treatment). Additionally, based on the most pronounced biological response to external stressors (see Table 1), two treatments will be selected for additional measurements ($n=6$ bionic corals per treatment). For all experiments nitro-oxidative stress will be determined as described in Aim 1 and Table 1.

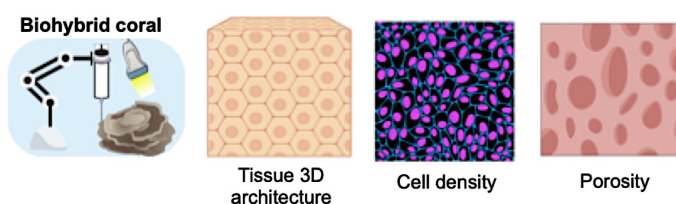


Figure 8: Aim 3 approach graphic representation

control conditions (see above, 25°C and downwelling irradiance= $300 \mu\text{mol photons m}^{-2} \text{s}^{-1}$, $n=6$ bionic corals per treatment). Additionally, based on the most pronounced biological response to external stressors (see Table 1), two treatments will be selected for additional measurements ($n=6$ bionic corals per treatment). For all experiments nitro-oxidative stress will be determined as described in Aim 1 and Table 1.

Aim 3.1. We will biofabricate a total of six different coral tissue host models that will be tested in pairs: 1,2) coral tentacles vs. coenosarc (i.e. connective tissues), 3,4) thick tissues (1mm) vs. thin tissues ($100 \mu\text{m}$), 5,6) oral epiderm vs. aboral epiderm. We have recently succeeded in developing a novel approach for the fabrication of different coral tissue types (oral epiderm, aboral endoderm and gastric cavities, Fig.

9) and we will now exploit this approach to investigate differences in nitro-oxidative stress. To create the different tissue types alginate-methacrylate (AlgMa) will be used to encapsulate *Breviolum minutum* at controlled volumetric and spatial densities (Cunning & Baker, 2012), while glycidyl methacrylate hyaluronic acid (GM-HA) will be used as sacrificial ink (Wangpraseurt et al., 2022). Tentacles and connective tissues will be bioprinted using our digital-light processing (DLP) based bioprinting approach according to the body plan of *Pocillopora acuta* (Wangpraseurt et al. 2020). DLP-based printing will also facilitate the engineering of different tissue thicknesses (100 μm vs 1mm) with micron-scale resolution. To create aboral and oral tissues, we will engineer gastric cavities using GM-HA (as described previously, Wangpraseurt et al., 2022). The exact thickness of the tissue layers and cavities will be determined via Optical Coherence tomography imaging of at least 5 coral specimens (see Wangpraseurt et al., 2017).

Aim 3.2. To test the role of symbiont density in modulating nitro-oxidative stress (e.g. (Cunning & Baker, 2012)) we will systematically vary the symbiont densities in our biohybrid coral models. One major advantage of the biohybrid coral model system is that we can easily manipulate symbiont densities by varying the pre-polymer ink composition (Wangpraseurt et al., 2022, You et al., 2023). Based on the results from Aim3.1. We will choose two tissue types that will be studied at a gradient of at least 5 symbiont densities ranging between low densities of $0.1 \times 10^5 \text{ cells cm}^{-2} \text{ s}^{-1}$ (representing bleached corals), and up to healthy densities of $\sim 3.5 \times 10^6 \text{ cells cm}^{-2} \text{ s}^{-1}$. For each density 6 replicate models will be printed.

Aim 3.3. To test the role of the coral skeleton in affecting nitro-oxidative stress from coral tissues, we will exploit our DLP-based bioprinting approach to systematically modulate skeletal microarchitecture. We recently developed a simple bioink glue combined with a post-polymerization technique to investigate differences in H_2O_2 production in tissues on top of a non-porous skeleton vs a porous skeleton (*Platygyra*, porosity =55.6 %, Wangpraseurt et al., 2022, Fig. 10). This preliminary data showed approximately 2-fold lower H_2O_2 production for porous *Platygyra* skeletons compared to solid non-porous skeletons (Fig. 10). We will now build upon this, to systematically investigate nitro-oxidative stress along a skeletal porosity gradient ranging from 0 to 80% porosity with at least 5 porosity values (0, 20, 40, 60, and 80%). The average skeletal pore size of *Pocillopora acuta* will be determined via OCT imaging (Wangpraseurt et al., 2017). To fabricate the multi-layer tissue construct with identical densities of *B. minutum*, we will attach different 3D printed skeletal structures to our tissue model using our post-polymerization method (Wangpraseurt et al., 2022, Fig. 10). For each porosity 6 replicate models will be printed.

Anticipated Outcomes:

We anticipate that the results of aim 3 will profoundly transform our understanding of the coral microenvironmental landscape in modulating nitro-oxidative stress. We hypothesize that microhabitats with rapid chemical gas exchange will negatively correlate with measurements of nitro-oxidative stress. Thus enhanced gas diffusion via the presence of pronounced gastric cavities and skeletal pores are expected to be an important missing link in understanding nitro-oxidative stress of whole coral colonies.

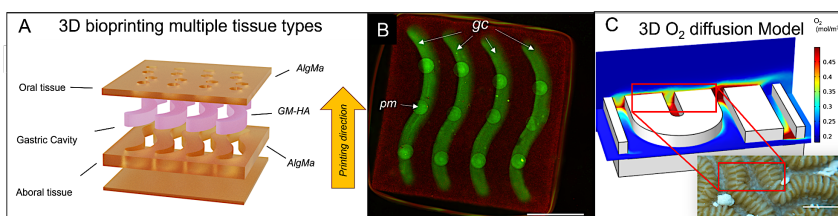


Figure 9. 3D bioprinting of multiple coral tissue types. A). Sketch visualizing our previous approach to generate gastric cavities and oral and Aboral coral tissue layers. B) Top view of successfully bioprinted gastric cavities (gc) and polyp mouth (pm) openings. C) In silico quantification of 3D O_2 gas exchange that is controlled by coral tissue microarchitecture. (Wangpraseurt et al., 2022)

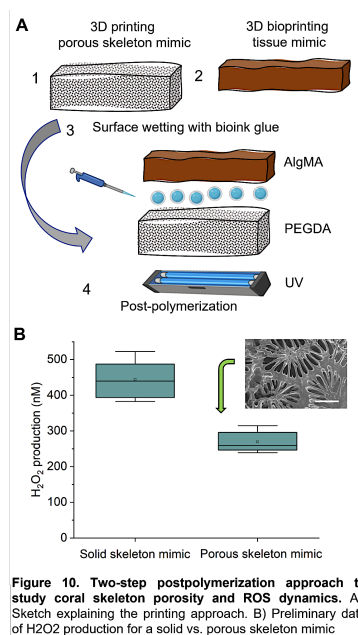


Figure 10. Two-step postpolymerization approach to study coral skeleton porosity and ROS dynamics. A). Sketch explaining the printing approach. B) Preliminary data of H_2O_2 production for a solid vs. porous skeleton mimic

These spatial patterns are likely further modulated by differences in symbiont cell densities and we expect to identify tipping points upon which an increase in cell density will non-linearly amplify ROS and RNS production of individual cells.

6. Milestones and Timeline, and Risk Assessment with Mitigation Plan

The detailed activities and their timelines are laid out in Fig. 11A. Risks to success, likelihood and mitigation plan are included here in Fig. 11B.

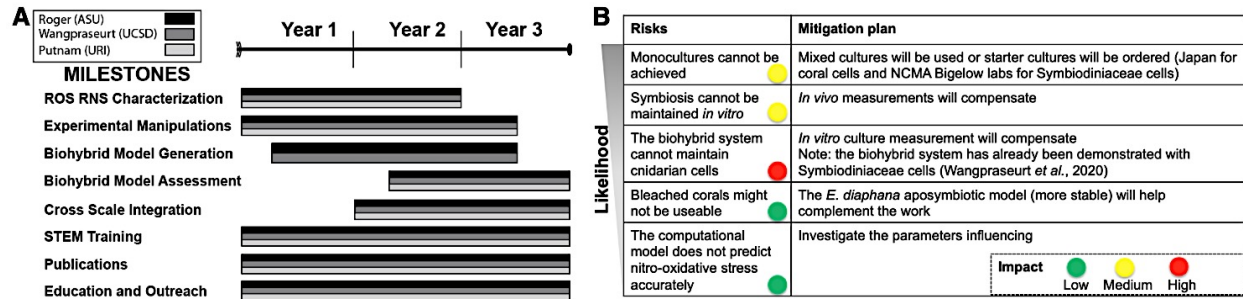


Figure 11: Project timeline including milestones (A) and risk assessment with mitigation plan (B)

7. Assessment of Experimental Goals

In order to assess the success of the project throughout its duration we have included goals, Key Performance Indicators (KPIs) and assessment metrics relevant to each Aim subAim in Table 3.

8. Broader Impacts

Component 1: Early Career Training in STEM: A major component of the broader impacts of the project will be focused on mentoring and training the next generation of scientists to think in multidisciplinary and integrative ways. To achieve the project's scientific and training goals, the research team will support at least 2 postdocs, 3 graduate, and 3 undergraduate students, providing a tiered training and mentoring system. In order to understand, integrate, and interpret the multifaceted datastreams, we will provide our students a comprehensive introduction to and training in the areas of biochemistry, bioengineering, cellular and molecular biology, and organismal physiology.

Component 2: Integration and Inclusion through scientific exchange and mentoring: A second component of our Broader Impact activities will emphasize diversity and inclusion within our integrative science training. Our project provides the essential context to generate the next generation of scientists with keen skills in critical thinking, data management, open and FAIR science, quantitative approaches, coding, publication, peer review, presentations, tiered mentoring, and public education and to do this with an inclusive lens. In order to achieve this, we will recruit diverse students and postdocs from historically marginalized groups through advertisement and mentoring investments in the following programs: URI

Table 3: Performance assessment

GOALS	KPIs	METRICS
1.1. Quantify ROS and RNS dynamics in cnidarian host cells in isolation	Intracellular concentration of different reactive oxygen species and different reactive nitrogen species in the systems and cells tested	•Number of <i>in vitro</i> assays using coral cells; •Number of <i>in vitro</i> assays using anemone cells; •Number of samples analyzed for gene sequences;
1.2. Quantify ROS and RNS dynamics in free-living Symbiodiniaceae		•Number of <i>in vitro</i> assays using Symbiodiniaceae cells; •Number of samples analyzed for gene sequences;
1.3. Quantify ROS and RNS dynamics in cnidarian host and Symbiodiniaceae in symbiosis		•Number of <i>in vitro</i> assays using symbiotic cells; •Number of corals analyzed; •Number of anemones analyzed; •Number of samples analyzed for gene sequences;
2.1. Characterize the response of the subcellular architecture to nitro-oxidative stress	3D view of organelles over time during nitro-oxidative stress	•Number of samples analyzed for 3D reconstruction; •Different cells 3D reconstituted;
2.2. Transcriptional expression of nitro-oxidative stress at the cellular level	Molecular cascade of nitro-oxidative stress	•Number of samples analyzed for gene sequences; •Different cells analyzed;
2.3. Define the boundaries of the antioxidant defense and tipping points	Computational model of nitro-oxidative stress in symbiotic systems	•Percent accuracy of model predictions; •Numbers of scenarios tested;
3.1. Characterize the response to nitro-oxidative stress in tissues of different architectures	Impact of tissue thickness and overall coral colony morphology on nitro-oxidative stress and free-radical diffusion	•Number of tissue thickness models tested; •Number of coral colony growth types tested;
3.2. Characterize the response to nitro-oxidative stress in tissues with different symbiont densities	Impact of algal cell density on nitro-oxidative stress and free-radical diffusion	•Range of cell density tested;

SACNAS chapter, ASU's NSF Louis Stokes Alliances for Minority Participation (LSAMP) Bridge to the Doctorate (BD) program, and UCSD's California Louis Stoke Alliance for Minority Participation (CAMP) program. Following recruitment in the project, we will focus on retention through the building of network collaboration and mentoring. These include for example: 1) postdoc led monthly all hands project meetings; 2) a distributed seminar with co-instruction across ASU, UCSD, URI; 3) cross institution tiered mentoring with pairing of postdocs and students connecting the teams.

Component 3: Open and Transparent Science: The third pillar of our Broader Impacts activities is further developing open science, research reproducibility, and quantitative skills of the scientific community. We train and work using open electronic lab notebooks, protocols, code repositories, and release raw data immediately following quality control. In this way we will facilitate the rapid advancement of scientific advancement and innovation and are aligned with the White House Office and Science and Technologies new policy on rapid data release. Importantly, our students and postdocs will be equipped for a future of faster and more efficient collaboration at the global level and at the intersection of multiple disciplines. The trainees will work specifically in approaches such as GitHub version control, Overleaf and Google doc shared manuscript writing, OpenScienceFramework large file archival, and R, Matlab, bash, and python programming.

Component 4: Public Outreach: The fourth component of our Broader Impacts is targeted to the broader public of all age ranges. **ASU:** In collaboration with students of ASU's Herberger Institute for Design and the Arts and the Arizona Science Museum, the ASU team will build an exhibition destined to showcase marine sciences, coral reefs and the research undertaken as part of the proposed project. This exhibition will leverage the recent launch of Pantone's Color of Biodiversity as a tool to introduce the people of Arizona to marine biodiversity, conservation and why, despite living in the Sonoran Desert, they should care about the ocean and coral reefs. The exhibition will be translated into Spanish to be accessible to the hispanic community. The ASU team will also work with Phoenix's Bioscience High School (BHS). This partnership will consist of students visiting the Marine Biochemistry Lab (Roger lab) at ASU and members of the ASU team visiting BHS to work on modules developed by the team focused on marine sciences and coral symbiosis. **UCSD:** We will partner with the Birch Aquarium to present our research to the public audience. At Birch, we plan to give public seminars at different events, incl. at "Science! Exploration! Adventure!" (SEA) days. SEA days will inform the general public about the broader impacts of our research through multimedia presentations and is an ideal vehicle for public engagement as these are established events with a large audience. We will also work with the Birch Aquarium at Scripps to expand current exhibitions and highlight the mechanisms underlying coral bleaching in interactive ways. For this, we plan to 3D print coral-like structures (using standard PLA printing) for educational purposes. Additionally, we are partnered with the Ocean Discovery Institute (ODI) which engages over 5,000 students, from the highest poverty and most diverse communities in the region, in rigorous science and conservation education. **URI:** Given the UNESCO's Greening Education Partnership and Public Radio KQUED's recent commentary on the role of schools in addressing climate change, the URI team will partner with the Synergy II project from the Art League RI that combines art and environmental and life sciences, as well as basic science research related to climate change and includes a particular focus on ocean science. Specifically, we will work with the Art League RI Synergy II project on artist-scientist exhibitions, as well as incorporating active learning opportunities for students and the public. This will consist of visual and performing arts projects focused on symbiosis and the ocean with K-12 schools across the state including: 1) Rocky Hill Country Day School (East Greenwich, Private); 2) Jacqueline M. Walsh School for the Performing and Visual Arts (Pawtucket, Charter); and 3) Providence Career and Technical Academy (Providence, Public).

Assessment of Broader Impacts

Success of our broader impact in STEM training will be assessed by quantifying the number of project trainees and pre/post surveys of multidisciplinary training activities and team equity and inclusion through collaborative science. Success in open data will be quantified as the number of data sets released, the

number of code releases, protocols and open lab notebook posts. For public outreach we will assess the number of outreach events organized, attendees to events, exhibit visitors at Birch Aquarium and Ocean Discovery Institute, and science art exhibitions, contact hours between the artists and science team in the Synergy project, number of media products generated (e.g. social media posts, news articles, videos), and public engagement with the media products (e.g., posts, likes, shares, views).

9. Results from Prior NSF Support

PI: Liza M. Roger (Arizona State University): No prior NSF support.

PI: Hollie Putnam (University of Rhode Island): (a) OCE-1756623, \$489,080, 06/15/18 - 05/31/22; (b) NSF-BSF: COLLABORATIVE RESEARCH: Elucidating adaptive potential through coral holobiont functional integration. (c) Accomplishments Intellectual Merit: The major goals were to characterize the holobiont stress response to acute and chronic stressors in environmentally resilient to susceptible coral taxa. This was accomplished through characterization of the dynamics of the coral microbiome, host transcriptome, holobiont metabolome, and epigenetic regulation across stressor treatments and time. Further, network analyses were used to elucidate the functional interactions between these players and physiology. High temperature had the greatest detrimental effects, with *Pocillopora acuta* showing much greater sensitivity through higher bleaching, lower growth, and lower survivorship than *Montipora capitata*. There were metabolites clearly diagnostic of thermal stress in both species. (c) Accomplishments Broader Impacts: We have presented our experiments and results from our research at a variety of public venues including: Mystic Aquarium Women in Science Day, and public presentations and Chasing Coral panel (Seattle Aquarium, Skyline High School, University House Retirement Community). Additionally, Putnam co-organized a session on Marine Environmental Epigenetics at ASLO 2019, a special issue of Marine Environmental Epigenetics in *Frontiers*, a session on Marine Environmental Epigenetics at the International Coral Reef Symposium, and the 2nd Congress on Epigenetics in Marine and Aquatic Research (EPIMAR). This project supported graduate and undergraduate students training, thesis, and dissertations. (d) Publications and Products: 11 peer reviewed publications (Cleves et al., 2020; Eirin-Lopez & Putnam, 2021; Innis et al., 2021; Putnam, 2021; {Shumaker et al., 2019; Stephens et al., 2021, 2022; Trigg et al., 2021; Van Etten et al., 2020; Williams, Chiles, et al., 2021; Williams, Pathmanathan, et al., 2021) and four coral genomes and two methylomes sequenced. Evidence of research data: Through this project we generated extensive full open access open science research products (e.g., Github data/code repositories, and electronic open and online lab notebooks).

PI: Daniel Wangpraseurt (University of California San Diego): (a) IOS - 2149925, \$630,129, 06/15/22 - 05/31/25; (b) NSF-BSF: Photophysiology and bio-optics of Red Sea mesophotic corals (c) Accomplishments Intellectual Merit: This project was just initiated in August 2022. The goals of our work are to: (1) analyze the photophysiology of corals along a depth-irradiance gradient, (2) conduct lab-based light microsensor measurements with concurrent endogenous green fluorescent protein and photosynthetic assays; (3) characterize tissue and skeletal morphology for the development of 3D light-capture models using optical coherence tomography and microcomputed tomography; and (4) apply our novel 3D bioprinting approach to experimentally determine the roles of coral skeleton morphology on photosynthesis. (c) Accomplishments Broader Impacts: Our broader impacts, still at early stages, will include museum exhibitions and the design of creative experiences for all ages. (d) Publications and Products: We have just initiated this project four months ago and have no publications to date.